

Reinforcement Learning from Human Feedback

A Beginner's Visual Guide — No Maths Required!

What is RLHF?

RLHF stands for **Reinforcement Learning from Human Feedback**. It is the technique used to train AI assistants like **ChatGPT** to be helpful, honest, and harmless.

Think of it like training a puppy:

■ Training a Puppy	■ Training an AI
You ask it to sit.	You give the AI a question.
It tries different behaviours.	It generates several possible answers.
You reward the good ones.	Humans pick the best answer.
It learns to sit reliably.	The AI learns to give great answers.

■ Key idea: Human feedback is the "teacher" that shapes how the AI behaves.

Why Do We Need RLHF?

Before RLHF, AI models were trained on huge amounts of internet text. They became very good at predicting the next word — but that also meant they sometimes generated harmful, rude, or nonsensical responses.

■ Problem (Before RLHF)	■ Solution (With RLHF)
AI could generate toxic or false content	Human raters flag bad outputs
AI didn't know what "helpful" means	Humans teach it by choosing better answers
AI optimised for word prediction only	AI optimises for human satisfaction

The 3 Steps of RLHF



Step 1 — Supervised Fine-Tuning (SFT)

Skilled human writers create ideal question-answer pairs. The base AI model is trained on these examples so it learns the right style, tone, and format for responses.

Step 2 — Reward Model Training

The AI generates several different answers to a question. Human raters rank those answers from best to worst. A second AI — the Reward Model — is trained to predict these human rankings so it can score any new answer automatically.

Step 3 — RL Fine-Tuning with PPO

The main AI generates answers. The Reward Model scores them. Using a technique called PPO (Proximal Policy Optimisation), the AI is nudged to produce answers that get higher scores —

essentially learning from a stream of automated human-like feedback.

Real-World Example: How ChatGPT Uses RLHF

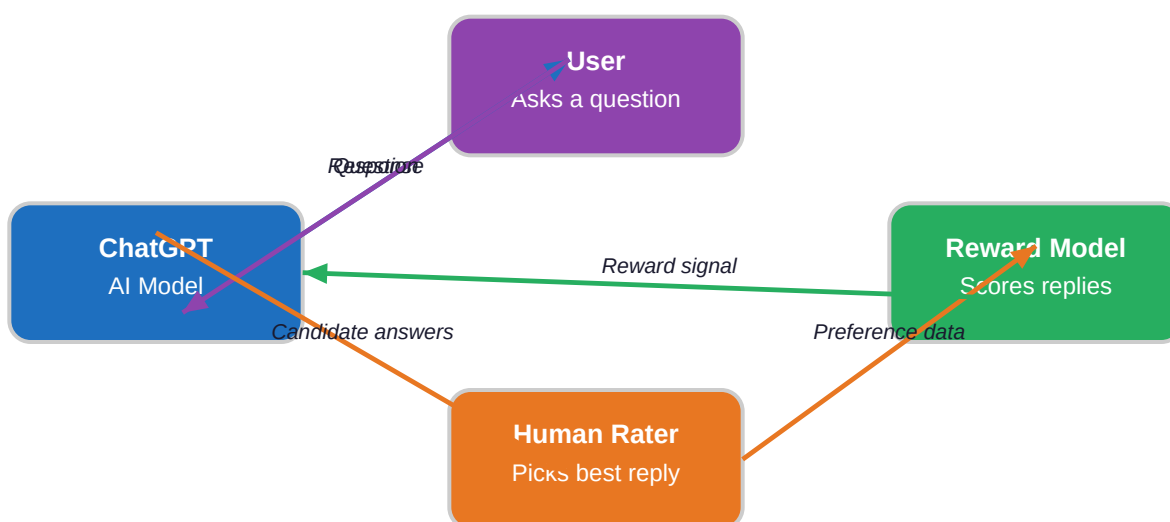
Let's walk through a concrete example. Imagine you type this into ChatGPT:

■ **"Explain black holes to a 10-year-old."**

Here is what happened behind the scenes during ChatGPT's training:

#	Who?	What happened?
1	OpenAI writers	Wrote ideal child-friendly explanations → used to train GPT initially.
2	ChatGPT (draft)	Generated 3 candidate answers: A) "A black hole is a singularity with infinite density..." B) "Imagine a giant vacuum cleaner in space!" C) A mix of jargon and simple words.
3	Human rater	Ranked: B > C > A (B is clearest for a 10-year-old).
4	Reward Model	Learned that simple, fun analogies score higher for kid-friendly questions.
5	PPO Training	ChatGPT is updated to favour simple, engaging answers like B in future.

The feedback loop visualised:



Over millions of such feedback cycles, ChatGPT became significantly better at giving helpful, accurate, and safe answers.

Key Takeaways

- RLHF teaches AI to be helpful, not just to predict text.
- Human raters are the secret ingredient — their preferences shape the AI.
- The Reward Model acts as an automated stand-in for human judgment.
- The cycle repeats thousands of times, gradually improving the AI.
- RLHF helps reduce harmful, biased, or misleading outputs.

Quick Glossary

Term	Plain-English Meaning
Reinforcement Learning (RL)	A type of AI training where the model learns by receiving rewards for good actions.
Human Feedback	Ratings or rankings provided by real people to guide the AI.
Reward Model	A helper AI that predicts how a human would rate a given answer.
PPO	Proximal Policy Optimisation — a math technique to safely update the AI's behaviour.
SFT	Supervised Fine-Tuning — training the AI on human-written examples first.
Policy	In RL, the "policy" is just the AI model deciding what answer to produce.